A Dual-Track Adaptive Projection Algorithm for a Nonlinear Split Feasibility Problem with Mixed Convex–Pseudoconvex Constraints

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ABSTRACT
We study a nonlinear split feasibility problem whose output set is a normal box and whose component functions are differentiable and each convex or pseudoconvex. We propose the Dual-Track Adaptive Projection (DTAP) algorithm, which splits the constraints into a convex and a pseudoconvex track and works on the nearly most violated constraints of each. DTAP combines a residual-scaled Polyak-type step on the convex track with a safeguarded diminishing step on the pseudoconvex track through a residual-responsive proportional weight, and then projects onto the base convex set. Its step sizes require no prescribed operator norm, Lipschitz constant, or Hölder modulus; the proof uses local bounds on the bounded iterate region. Under consistency and continuous differentiability, we prove that the whole sequence converges strongly to a feasible point. As an application, we formulate feature selection as such a problem, with constraints built from fractional and fuzzy-rough feature-selection criteria. Experiments on UCI datasets show that the selected features keep the classification accuracy close to that of the full feature set, and test problems with a nonconvex pseudoconvex constraint exercise the pseudoconvex track; we also compare DTAP with subgradient methods and study its sensitivity to the starting point and the step scale.

KEYWORDS
Split feasibility problem; Pseudoconvex optimization; Tchebyshev scalarization; Adaptive step size; Strong convergence; Feature selection

AMS CLASSIFICATION
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1. Introduction

The split feasibility problem (SFP) was introduced by Censor and Elfving [4] as a model for constrained inverse problems. Given nonempty closed convex sets $C \subseteq \mathbb{R}^n$ and $Q \subseteq \mathbb{R}^m$ and a matrix $\mathcal{A} \in \mathbb{R}^{m\times n}$, it asks to

$$\text{find } x^\star \in C \quad \text{such that} \quad \mathcal{A}x^\star \in Q. \tag{1}$$

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Byrne's CQ algorithm [3] iterates

$$x^{k+1} = P_C\big(x^k - \tau \mathcal{A}^\top (I - P_Q)\mathcal{A}x^k\big), \qquad 0 < \tau < 2/\left\|\mathcal{A}\right\|^2. \tag{2}$$

Here $\|\mathcal{A}\|$ is the spectral norm of $\mathcal{A}$. Relaxed variants project onto supporting half-spaces [17]; self-adaptive steps avoid estimating $\|\mathcal{A}\|$ [10, 20, 22]; operator splitting [21] and alternating direction methods [18] decompose the problem. These ideas are standard; DTAP adapts them to the nonlinear case.

Replacing $\mathcal{A}$ by a nonlinear map F gives the nonlinear SFP [9, 19]

$$\text{find } x^\star \in K \quad \text{such that} \quad F(x^\star) \in Q, \tag{3}$$

where $K \subseteq \mathbb{R}^n$ is nonempty, closed and convex, $Q = \prod_{i=1}^m (-\infty, b_i]$ is a normal box, and each F_i is differentiable and convex or pseudoconvex on K. Since pseudoconvex functions have convex sublevel sets, the feasible set $\Omega = \{x \in K : F(x) \in Q\}$ is convex. We seek a point of Ω, not a minimizer of any F_i.

Li et al. [9] assume a convex aggregate proximity function, which componentwise pseudoconvexity does not ensure. Yen et al. [19] use the same max-type violation measure and active-set device, but require quasilinear components, a class incomparable with ours. Hu et al. [7] treat quasiconvex inequalities, which contain our class, without using the gradient inequality of convex components.

Because Q is a box, the ℓ_∞-distance from $F(x)$ to Q has the closed form

$$J(x) = d_\infty(F(x), Q) = \max_{1 \le i \le m} (F_i(x) - b_i)^+.$$

DTAP splits the constraints into a convex track and a pseudoconvex track. On each track it sums the gradients of the nearly most violated constraints, weighted by their violations. The gradient inequality of convex functions justifies a residual-scaled Polyak-type step on the convex track [10, 14]; pseudoconvexity gives only the sign of a directional derivative, so the pseudoconvex track uses a safeguarded diminishing step. A proportional weight combines the two steps, and the result is projected onto K; the proof uses this weight only to keep a persistently violated track active (Remark 1). No prescribed operator norm, Lipschitz constant, or Hölder modulus is needed: a quasi-Fejér inequality shows that the iterates are bounded, and the proof then uses local bounds, without Lipschitz gradients.

As an application, upper bounds on three criteria from a fractional model [15] and a fuzzy-rough model [16] define the constraints; a feasible weight vector satisfies all three bounds but need not solve either source problem. These constraints are convex (Proposition 4.1), so we also use test problems with a nonconvex pseudoconvex constraint to exercise the pseudoconvex track.

The contributions are as follows.

C1. A dual-track method that uses the gradient inequality on the convex constraints and only sign information on the pseudoconvex ones, without assuming a convex proximity function, unlike [9], or a Hölder modulus, unlike the dynamic step of [7].

C2. Convergence of the whole sequence to a feasible point under Assumption 1 (Theorem 3.12).

C3. Experiments that report constraint activity, compare DTAP with tuned subgradient methods, and measure its sensitivity to the starting point and the step

scale.

Section 2 gives preliminaries. Section 3 presents DTAP and its convergence analysis. Section 4 reports experiments, and Section 5 concludes.

2. Preliminaries

Throughout, $H = \mathbb{R}^n$ is the Euclidean space with inner product $\langle \cdot, \cdot \rangle$ and induced norm $\|\cdot\|$. All convergence statements concern the norm topology. Bounded sequences have convergent subsequences. The standard projection and generalized convexity facts below can be found in [1, 12].

For a nonempty closed convex set $K \subseteq H$, the metric projection $P_K(x)$ is the unique minimizer of $\|y - x\|$ over $y \in K$.

Proposition 2.1. *For $x \in H$ and $p \in K$, $p = P_K(x)$ if and only if*

$$\langle x - p,\, y - p \rangle \le 0 \qquad \text{for every } y \in K. \tag{4}$$

This characterization yields Lemma 3.2 below.

A function $f : K \to \mathbb{R}$ is convex if $f((1-t)x + ty) \le (1-t)f(x) + tf(y)$ for $x, y \in K$ and $t \in [0, 1]$. The following weaker derivative condition was introduced by Mangasarian [11].

Definition 2.2. A function f that is differentiable on an open neighbourhood of K is pseudoconvex on K if, for all $x, y \in K$,

$$f(y) < f(x) \quad \Longrightarrow \quad \langle \nabla f(x),\, y - x \rangle < 0. \tag{5}$$

Every differentiable convex function is pseudoconvex. Pseudoconvex functions have convex sublevel sets, although the functions themselves need not be convex.

3. The DTAP algorithm and its convergence analysis

We formulate the normal-box NSFP, state DTAP, and prove convergence of the whole sequence to a feasible point.

3.1. Problem statement and Tchebyshev scalarization

Let $K \subseteq H$ be nonempty closed convex, and let $U \subseteq H$ be an open neighbourhood of K. Fix an integer $m \ge 1$, let $F = (F_1, \dots, F_m) : U \to \mathbb{R}^m$, let $b = (b_1, \dots, b_m) \in \mathbb{R}^m$, and set the normal box $Q := (-\infty, b_1] \times \cdots \times (-\infty, b_m]$. We study the instance of (3)

$$\text{find } x^\star \in \Omega, \qquad \Omega := \big\{ x \in K : F_i(x) \le b_i,\ i = 1, \dots, m \big\}, \tag{6}$$

and call Ω the *feasible set*. To turn $F(x) \in Q$ into a tractable merit function, we measure the residual in the sup-norm. For $z = F(x)$, $d_\infty(z, Q) = \inf_{q \le b} \max_{1 \le i \le m} |z_i - q_i|$. By the box structure the infimum separates over coordinates: if $z_i \le b_i$ then $\inf_{q_i \le b_i} |z_i - q_i| = 0$ (at $q_i = z_i$), and if $z_i > b_i$ then it equals $z_i - b_i$ (at $q_i = b_i$).

Hence $\inf_{q_i \le b_i} |z_i - q_i| = (z_i - b_i)^+$ with $(t)^+ := \max\{t, 0\}$, so

$$d_\infty(z, Q) = \max_{1 \le i \le m} (z_i - b_i)^+. \tag{7}$$

Define the *error function*

$$J(x) := d_\infty\big(F(x), Q\big) = \max_{1 \le i \le m} \big(F_i(x) - b_i\big)^+. \tag{8}$$

Then $x \in K$ is feasible if and only if $J(x) = 0$; when $\Omega \neq \emptyset$, the feasible points are the minimizers of J over K. Formula (8) gives a closed-form feasibility residual whose maximizers identify the most violated constraints. Section 3.2 introduces an approximate active set to handle the nonsmooth maximum.

3.2. Decomposition into two tracks

We split $I := \{1, \dots, m\}$ into a *convex track* I^{cvx} and a *pseudoconvex track* I^{ps}, each with its own step rule. The convex track uses a Polyak step [14]; the pseudoconvex track uses a safeguarded diminishing step. A proportional weight combines the two updates.

Fix a parameter $\delta > 0$. For $x \in U$ define the componentwise violations

$$v_i(x) := \big(F_i(x) - b_i\big)^+, \tag{9}$$

so that $J(x) = \max_i v_i(x)$ by (8). Assumption 1 makes the partition precise. The *per-track maximal violations* are

$$J^{\text{cvx}}(x) := \max_{i \in I^{\text{cvx}}} v_i(x) \quad (=0 \text{ if } I^{\text{cvx}} = \emptyset), \tag{10}$$

$$J^{\text{ps}}(x) := \max_{j \in I^{\text{ps}}} v_j(x) \quad (=0 \text{ if } I^{\text{ps}} = \emptyset), \tag{11}$$

so $J(x) = \max\{J^{\text{cvx}}(x), J^{\text{ps}}(x)\}$. To cope with the nonsmoothness of the maximum, we replace the set of indices at which the maximum is attained by a δ-enlarged *approximate active set* on each track ('active' meaning within δ of the largest violation):

$$K^{\text{cvx}}(x) := \begin{cases} \{i \in I^{\text{cvx}} : v_i(x) \ge J^{\text{cvx}}(x) - \delta\} & \text{if } J^{\text{cvx}}(x) > 0, \\ \emptyset & \text{if } J^{\text{cvx}}(x) = 0, \end{cases} \tag{12}$$

$$K^{\text{ps}}(x) := \begin{cases} \{j \in I^{\text{ps}} : v_j(x) \ge J^{\text{ps}}(x) - \delta\} & \text{if } J^{\text{ps}}(x) > 0, \\ \emptyset & \text{if } J^{\text{ps}}(x) = 0. \end{cases} \tag{13}$$

On these sets we form the *per-track local functions and directions*

$$\Phi^{\text{cvx}}(x) := \tfrac{1}{2} \sum_{i \in K^{\text{cvx}}(x)} v_i(x)^2, \qquad g^{\text{cvx}}(x) := \sum_{i \in K^{\text{cvx}}(x)} v_i(x) \nabla F_i(x), \tag{14}$$

$$g^{\text{ps}}(x) := \sum_{j \in K^{\text{ps}}(x)} v_j(x) \nabla F_j(x). \tag{15}$$

The vectors $g^{\mathrm{cvx}}(x), g^{\mathrm{ps}}(x)$ are the two search directions, and $\Phi^{\mathrm{cvx}}(x)$ supplies the Polyak numerator on the convex track. Along the iterates we abbreviate $J_k^{\mathrm{cvx}} := J^{\mathrm{cvx}}(x^k)$, $g_k^{\mathrm{cvx}} := g^{\mathrm{cvx}}(x^k)$, $\Phi_k^{\mathrm{cvx}} := \Phi^{\mathrm{cvx}}(x^k)$, and likewise on the pseudoconvex track.

3.3. *Assumptions*

Assumption 1. We assume (A1)–(A3):

(A1) (a) $\Omega \neq \emptyset$ (consistency). (b) $I = \{1, \dots, m\}$ is partitioned as $I = I^{\mathrm{cvx}} \cup I^{\mathrm{ps}}$; each $F_i : U \to \mathbb{R}$ is convex on K if $i \in I^{\mathrm{cvx}}$ and pseudoconvex on K if $i \in I^{\mathrm{ps}}$.

(A2) Each $F_i : U \to \mathbb{R}$ is continuously differentiable on U.

(A3) (i) $\delta > 0$ fixed. (ii) $\rho_{\mathrm{cvx}} \in (0, 2)$. (iii) $\{\gamma_k\} \subset (0, \infty)$ with $\sum_{k=0}^{\infty} \gamma_k^2 < \infty$ and $\sum_{k=0}^{\infty} \gamma_k = \infty$.

The open-neighbourhood formulation permits fractional functions whose denominators are positive on K; the algorithm evaluates the functions and gradients only at points of K. Although some component functions may be nonconvex, Ω is convex: differentiable pseudoconvex functions are quasiconvex on K, so all constraint sublevel sets in K are convex.

Example 3.1. The sequence $\gamma_k := 1/(k+1)$ satisfies (A3)(iii), since $\sum_{k\geq 0}(k+1)^{-2} < \infty$ while $\sum_{k\geq 0}(k+1)^{-1} = \infty$. More generally $\gamma_k = (k+c)^{-s}$ with $c > 0$ and $s \in (1/2, 1]$ qualifies; the experiments use $s = 0.55$.

3.4. *The DTAP algorithm*

The Dual-Track Adaptive Projection algorithm combines the two track-specific step sizes through the proportional weight μ_k and projects the combined update onto the base convex set K. It is stated as Algorithm 1.

Algorithm 1 Dual-Track Adaptive Projection (DTAP) for the normal-box NSFP (6).

Input: map $F = (F_1, \dots, F_m)$, thresholds b, base set K, index split $I^{\mathrm{cvx}} \cup I^{\mathrm{ps}}$.
Parameters: $\delta > 0$, tolerance $\varepsilon > 0$, $\rho_{\mathrm{cvx}} \in (0, 2)$, sequence $\{\gamma_k\}$ satisfying (A3)(iii).
Initialization: choose $x^0 \in K$; set $k \leftarrow 0$.

Step 1 (per-track quantities). If $J(x^k) \leq \varepsilon$, stop and return x^k; otherwise:

(1a) Compute $v_i(x^k)$ for all i, and $J_k^{\mathrm{cvx}} = \max_{i \in I^{\mathrm{cvx}}} v_i(x^k)$, $J_k^{\mathrm{ps}} = \max_{j \in I^{\mathrm{ps}}} v_j(x^k)$ (each 0 if its track is empty).

(1b) Form the approximate active sets $K_k^{\mathrm{cvx}}, K_k^{\mathrm{ps}}$ via (12)–(13).

(1c) Form Φ_k^{cvx}, g_k^{cvx}, g_k^{ps} via (14)–(15).

(1d) Per-track step sizes (Polyak on the convex track, safeguarded on the pseudoconvex track):

$$\eta_k^{\mathrm{cvx}} = \begin{cases} \rho_{\mathrm{cvx}} \dfrac{2\Phi_k^{\mathrm{cvx}}}{\|g_k^{\mathrm{cvx}}\|^2} & \text{if } \|g_k^{\mathrm{cvx}}\| > 0, \\ 0 & \text{otherwise,} \end{cases} \qquad \eta_k^{\mathrm{ps}} = \frac{\gamma_k}{\max\{1, \|g_k^{\mathrm{ps}}\|\}}.$$

(1e) Proportional weight: $\mu_k = \dfrac{J_k^{\mathrm{cvx}}}{J_k^{\mathrm{cvx}} + J_k^{\mathrm{ps}}}$ if $J_k^{\mathrm{cvx}} + J_k^{\mathrm{ps}} > 0$, and $\mu_k = 1/2$ otherwise.

(1f) Combine the two directions: $y^k = x^k - [\mu_k \eta_k^{\mathrm{cvx}} g_k^{\mathrm{cvx}} + (1 - \mu_k)\eta_k^{\mathrm{ps}} g_k^{\mathrm{ps}}]$.

Step 2 (projection). $x^{k+1} = P_K(y^k)$; set $k \leftarrow k + 1$ and go to Step 1.

Remark 1. The proportional weight is an adaptive allocation based on the two maximal violations; we do not claim that it is the only choice for which DTAP converges, or that it is optimal. Its specific form is used in (27) to control the convex-track residual and in Claim 2 of Stage 2 of Theorem 3.12 to ensure that a persistently violated pseudoconvex track retains positive weight. Although the one-step inequality of Lemma 3.8 holds for every $\mu_k \in [0,1]$, it does not by itself imply convergence for every weight sequence.

Remark 2. The iteration does not require prescribed operator norms, Lipschitz constants, or Hölder moduli. The proof nevertheless uses finite bounds on function values and gradients on a compact set containing the iterates, after boundedness has been proved. Bounded gradients imply Lipschitz continuity of the functions on that convex compact set, but these bounds are not algorithm inputs. Assumption (A2) does not imply Lipschitz continuity of the gradients, and the theorem does not require it. The user still chooses ρ_{cvx}, δ, and $\{\gamma_k\}$; these parameters do not depend on any constant of the problem.

For the analysis we write the combined increment as

$$d_k := \mu_k \eta_k^{\mathrm{cvx}} g_k^{\mathrm{cvx}} + (1-\mu_k)\eta_k^{\mathrm{ps}} g_k^{\mathrm{ps}}, \tag{16}$$

so $y^k = x^k - d_k$ and $x^{k+1} = P_K(x^k - d_k)$.

3.5. Per-iteration cost

Each DTAP iteration evaluates the componentwise violations $v_i(x^k)$ and the active-set gradients $\{\nabla F_i(x^k) : i \in K_k^{\mathrm{cvx}} \cup K_k^{\mathrm{ps}}\}$, forms the two search directions $g_k^{\mathrm{cvx}}, g_k^{\mathrm{ps}}$ as weighted sums of these gradients, computes the step sizes, and performs one projection onto K. It uses neither an operator-norm estimate nor a line search. The cost is dominated by the constraint evaluations and the projection. The cost for the feature-selection instance is given in Section 4.1 (Remark 6).

3.6. Auxiliary lemmas

The two tracks need different arguments. On the convex track, Lemma 3.4 bounds $\langle \nabla F_i(x),\, x - x^\star\rangle$ from below by the violation $v_i(x)$; this gives a decrease of $\|x^k - x^\star\|^2$ of order $\mu_k T_k$. On the pseudoconvex track, definition (5) gives only the sign of this inner product. Lemma 3.8 combines both tracks in one quasi-Fejér inequality; the choice of the parameter ζ in Young's inequality keeps the convex-track term negative. Stage 2 of Theorem 3.12 then shows, by contradiction, that at least one cluster point is feasible, and Stage 3 shows that the whole sequence converges strongly to it. The two degenerate cases follow as corollaries.

Throughout, $\{x^k\}$ is the infinite sequence generated by Algorithm 1 with the stopping test removed. As shown below (Lemma 3.10 and Theorem 3.12), $J(x^k) \to 0$; hence for any $\varepsilon > 0$ there is a finite k with $J(x^k) \le \varepsilon$, so the finite-stopping version terminates after finitely many steps with an ε-feasible point.

We begin with two preparatory results used repeatedly below.

Lemma 3.2. *Let $K \subseteq H$ be nonempty closed convex. For every $y \in H$ and $z \in K$,*

$$\|P_K(y) - z\|^2 \le \|y - z\|^2 - \|y - P_K(y)\|^2 \le \|y - z\|^2 . \tag{17}$$

Proof. Put $p := P_K(y)$. By Proposition 2.1, $\langle y - p,\, z - p\rangle \le 0$, so expanding $\|y - z\|^2 = \|y - p\|^2 + 2\,\langle y - p,\, p - z\rangle + \|p - z\|^2$ and using $\langle y - p,\, p - z\rangle \ge 0$ gives $\|y - z\|^2 \ge \|y - p\|^2 + \|p - z\|^2$, which is (17). □

Lemma 3.3. *Let $\{a_k\} \subset [0, \infty)$ and $\{\varepsilon_k\} \subset [0, \infty)$ satisfy $a_{k+1} \le a_k + \varepsilon_k$ for all k and $\sum_{k=0}^{\infty} \varepsilon_k < \infty$. Then $\{a_k\}$ converges to a finite limit.*

Proof. $b_k := a_k - \sum_{l<k} \varepsilon_l$ is nonincreasing (from $a_{k+1} \le a_k + \varepsilon_k$) and bounded below by $-\sum_l \varepsilon_l > -\infty$, hence converges; since $\sum_{l<k} \varepsilon_l \to S < \infty$, $a_k = b_k + \sum_{l<k} \varepsilon_l$ converges. □

On the convex track, convexity gives a quantitative link between the gradient and the violation.

Lemma 3.4. *For $x^\star \in \Omega$, $i \in I^{\mathrm{cvx}}$, and $x \in K$,*

$$v_i(x)\ \langle \nabla F_i(x),\, x - x^\star\rangle \ge v_i(x)^2.$$

Proof. If $v_i(x) = 0$ both sides vanish. Let $v_i(x) > 0$, i.e. $F_i(x) > b_i$. Define $\varphi : [0, 1] \to \mathbb{R}$ by $\varphi(t) := F_i((1 - t)x + tx^\star)$; since K is convex, $[x, x^\star] \subset K$, so φ is well defined, and by (A1)(b) it is convex, while by (A2) it is differentiable with $\varphi'(0^+) = \langle \nabla F_i(x),\, x^\star - x\rangle$. The one-dimensional gradient inequality gives $\varphi(1) \ge \varphi(0) + \varphi'(0^+)$, i.e. $F_i(x^\star) \ge F_i(x) + \langle \nabla F_i(x),\, x^\star - x\rangle$. As $x^\star \in \Omega$ we have $F_i(x^\star) \le b_i < F_i(x)$, whence $\langle \nabla F_i(x),\, x - x^\star\rangle \ge F_i(x) - b_i = v_i(x)$. Multiplying by $v_i(x) > 0$ gives the claim. □

Remark 3. Summing Lemma 3.4 over the approximate active set K_k^{cvx} yields $\langle g_k^{\mathrm{cvx}},\, x^k - x^\star\rangle \ge \sum_{i \in K_k^{\mathrm{cvx}}} v_i(x^k)^2 = 2\Phi_k^{\mathrm{cvx}}$, a bound used in the proofs of Lemmas 3.8 and 3.10.

On the pseudoconvex track, only a sign is available.

Lemma 3.5. *For $x^\star \in \Omega$, $j \in I^{\mathrm{ps}}$, and $x \in K$, $v_j(x)\ \langle \nabla F_j(x),\, x - x^\star\rangle \ge 0$, with strict inequality when $v_j(x) > 0$.*

Proof. If $v_j(x) = 0$ both sides vanish. If $v_j(x) > 0$ then $F_j(x) > b_j \ge F_j(x^\star)$. By (A1)(b), $F_j|_K$ is pseudoconvex; applying (5) with $y = x^\star$ gives $\langle \nabla F_j(x),\, x^\star - x\rangle < 0$, i.e. $\langle \nabla F_j(x),\, x - x^\star\rangle > 0$. Multiplying by $v_j(x) > 0$ yields the strict inequality. □

Finally, two compactness-based norm bounds on the search directions.

Lemma 3.6. *Assume $I^{\mathrm{cvx}} \ne \emptyset$ and let $B \subset U$ be compact. With $L_B := \max_{i \in I^{\mathrm{cvx}}} \sup_{x \in B} \|\nabla F_i(x)\|$ (finite by (A2)), for every $x \in B$,*

$$\|g^{\mathrm{cvx}}(x)\|^2 \le 2\ |I^{\mathrm{cvx}}|\ L_B^2\, \Phi^{\mathrm{cvx}}(x).$$

Proof. Put $K' = K^{\mathrm{cvx}}(x)$, $m' = |K'| \le |I^{\mathrm{cvx}}|$. By the triangle and Cauchy–Schwarz

inequalities,

$$\|g^{\mathrm{cvx}}(x)\| = \Big\| \sum_{i\in K'} v_i(x)\nabla F_i(x) \Big\| \le \Big(\sum_{i\in K'} v_i(x)^2 \Big)^{1/2} \Big(\sum_{i\in K'} \|\nabla F_i(x)\|^2 \Big)^{1/2}.$$

Since $\sum_{i\in K'} v_i(x)^2 = 2\Phi^{\mathrm{cvx}}(x)$ and $\sum_{i\in K'} \|\nabla F_i(x)\|^2 \le m' L_B^2 \le |I^{\mathrm{cvx}}|\, L_B^2$, squaring gives the claim. □

Lemma 3.7. *Assume $I^{\mathrm{ps}} \neq \emptyset$ and let $B \subset U$ be compact. With $L'_B := \max_{j\in I^{\mathrm{ps}}} \sup_{x\in B} \|\nabla F_j(x)\|$ and $V_B := \max_{j\in I^{\mathrm{ps}}} \sup_{x\in B} v_j(x)$ (both finite by (A2)), for every $x \in B$,*

$$\|g^{\mathrm{ps}}(x)\| \le |I^{\mathrm{ps}}|\; V_B\, L'_B.$$

Proof. By the triangle inequality, $\|g^{\mathrm{ps}}(x)\| = \big\|\sum_{j\in K^{\mathrm{ps}}(x)} v_j(x)\nabla F_j(x)\big\| \le \sum_{j\in K^{\mathrm{ps}}(x)} v_j(x)\,\|\nabla F_j(x)\| \le |K^{\mathrm{ps}}(x)|\, V_B L'_B \le |I^{\mathrm{ps}}|\, V_B L'_B$. □

3.7. Boundedness and decay of the iteration error

The following inequality is of quasi-Fejér type in the sense of Combettes [5]. It yields boundedness of $\{x^k\}$ and decay of the convex-track violation. Set

$$c_0 := \frac{\rho_{\mathrm{cvx}}(2-\rho_{\mathrm{cvx}})}{2} \in \Big(0, \tfrac{1}{2}\Big], \qquad C_0 := \frac{2+\rho_{\mathrm{cvx}}}{2-\rho_{\mathrm{cvx}}} \in (1,\infty), \tag{18}$$

and

$$T_k := \begin{cases} \dfrac{(2\Phi_k^{\mathrm{cvx}})^2}{\big\|g_k^{\mathrm{cvx}}\big\|^2} & \text{if } \|g_k^{\mathrm{cvx}}\| > 0, \\ 0 & \text{if } \|g_k^{\mathrm{cvx}}\| = 0. \end{cases} \tag{19}$$

For fixed $x^\star \in \Omega$ write $R_k^{x^\star} := \langle g_k^{\mathrm{ps}},\, x^k - x^\star\rangle$.

Lemma 3.8. *For every $x^\star \in \Omega$ and $k \ge 0$,*

$$\big\|x^{k+1} - x^\star\big\|^2 \le \big\|x^k - x^\star\big\|^2 - c_0\mu_k T_k - 2(1-\mu_k)\eta_k^{\mathrm{ps}} R_k^{x^\star} + C_0\gamma_k^2. \tag{20}$$

Proof. Apply Lemma 3.2 with $y = y^k$ and $z = x^\star \in K$ and use $y^k = x^k - d_k$ from (16):

$$\big\|x^{k+1} - x^\star\big\|^2 \le \big\|x^k - x^\star\big\|^2 - 2\left\langle d_k,\, x^k - x^\star\right\rangle + \|d_k\|^2. \tag{21}$$

Cross term. Expanding d_k,

$$\left\langle d_k,\, x^k - x^\star\right\rangle = \mu_k\eta_k^{\mathrm{cvx}} \left\langle g_k^{\mathrm{cvx}},\, x^k - x^\star\right\rangle + (1-\mu_k)\eta_k^{\mathrm{ps}} R_k^{x^\star}. \tag{22}$$

If $\|g_k^{\mathrm{cvx}}\| = 0$ then $\eta_k^{\mathrm{cvx}} = 0$ and the first term vanishes. If $\|g_k^{\mathrm{cvx}}\| > 0$, Remark 3 gives $\langle g_k^{\mathrm{cvx}},\, x^k - x^\star\rangle \ge 2\Phi_k^{\mathrm{cvx}}$; multiplying by $\mu_k\eta_k^{\mathrm{cvx}} \ge 0$ and substituting $\eta_k^{\mathrm{cvx}} =$

$2\rho_{\mathrm{cvx}}\Phi_k^{\mathrm{cvx}}/\left\|g_k^{\mathrm{cvx}}\right\|^2$,

$$\mu_k\eta_k^{\mathrm{cvx}}\left\langle g_k^{\mathrm{cvx}},\, x^k-x^\star\right\rangle \geq \mu_k\rho_{\mathrm{cvx}}\frac{(2\Phi_k^{\mathrm{cvx}})^2}{\left\|g_k^{\mathrm{cvx}}\right\|^2}=\mu_k\rho_{\mathrm{cvx}}T_k. \tag{23}$$

Quadratic term. By Young's inequality $\|u+v\|^2\leq(1+\zeta)\|u\|^2+(1+\zeta^{-1})\|v\|^2$ with $u=\mu_k\eta_k^{\mathrm{cvx}}g_k^{\mathrm{cvx}}$, $v=(1-\mu_k)\eta_k^{\mathrm{ps}}g_k^{\mathrm{ps}}$,

$$\|d_k\|^2\leq(1+\zeta)\rho_{\mathrm{cvx}}^2\mu_k^2T_k+(1+\zeta^{-1})(1-\mu_k)^2(\eta_k^{\mathrm{ps}})^2\left\|g_k^{\mathrm{ps}}\right\|^2.$$

By definition $\eta_k^{\mathrm{ps}}=\gamma_k/\max\{1,\|g_k^{\mathrm{ps}}\|\}$, so $\eta_k^{\mathrm{ps}}\|g_k^{\mathrm{ps}}\|\leq\gamma_k$ and hence $(\eta_k^{\mathrm{ps}})^2\|g_k^{\mathrm{ps}}\|^2\leq\gamma_k^2$; with $(1-\mu_k)^2\leq1$,

$$\|d_k\|^2\leq(1+\zeta)\rho_{\mathrm{cvx}}^2\mu_k^2T_k+(1+\zeta^{-1})\gamma_k^2. \tag{24}$$

Choose $\zeta:=(2-\rho_{\mathrm{cvx}})/(2\rho_{\mathrm{cvx}})>0$. Then $(1+\zeta)\rho_{\mathrm{cvx}}=(2+\rho_{\mathrm{cvx}})/2$, so $(1+\zeta)\rho_{\mathrm{cvx}}^2=\rho_{\mathrm{cvx}}(2+\rho_{\mathrm{cvx}})/2$, and $1+\zeta^{-1}=(2+\rho_{\mathrm{cvx}})/(2-\rho_{\mathrm{cvx}})=C_0$. Combining (21), (22), (23) and (24),

$$\left\|x^{k+1}-x^\star\right\|^2\leq\left\|x^k-x^\star\right\|^2+\left[-2\mu_k\rho_{\mathrm{cvx}}+(1+\zeta)\rho_{\mathrm{cvx}}^2\mu_k^2\right]T_k-2(1-\mu_k)\eta_k^{\mathrm{ps}}R_k^{x^\star}+C_0\gamma_k^2.$$

The bracket equals $\mu_k\rho_{\mathrm{cvx}}\big(-2+\frac{2+\rho_{\mathrm{cvx}}}{2}\mu_k\big)$. Since $\rho_{\mathrm{cvx}}<2$, for all $\mu_k\in[0,1]$ the factor $-2+\frac{2+\rho_{\mathrm{cvx}}}{2}\mu_k$ is at most $-(2-\rho_{\mathrm{cvx}})/2<0$ (its maximum over $[0,1]$ is at $\mu_k=1$); multiplying by $\mu_k\rho_{\mathrm{cvx}}\geq0$ gives the bracket $\leq-c_0\mu_k$. This yields (20). □

Lemma 3.9. *The sequence $\{x^k\}$ is bounded. Moreover, for each fixed $x^\star\in\Omega$:* (i) $\{\|x^k-x^\star\|^2\}$ *converges in* $[0,\infty)$*;* (ii) $\sum_{k=0}^\infty\mu_kT_k<\infty$*;* (iii) $\sum_{k=0}^\infty(1-\mu_k)\eta_k^{\mathrm{ps}}R_k^{x^\star}<\infty$.

Proof. Fix $x^\star\in\Omega$ and set $a_k:=\|x^k-x^\star\|^2$. By Lemma 3.5, $R_k^{x^\star}\geq0$, and $\mu_k,T_k,(1-\mu_k)\eta_k^{\mathrm{ps}}\geq0$, so (20) gives $a_{k+1}\leq a_k+C_0\gamma_k^2$. Since $\sum_kC_0\gamma_k^2<\infty$ by (A3)(iii), Lemma 3.3 yields (i), and boundedness of $\{x^k\}$ follows. Summing (20) from 0 to $N-1$,

$$a_N+\sum_{k=0}^{N-1}\big(c_0\mu_kT_k+2(1-\mu_k)\eta_k^{\mathrm{ps}}R_k^{x^\star}\big)\leq a_0+C_0\sum_{k=0}^{N-1}\gamma_k^2.$$

The right side is bounded uniformly in N; letting $N\to\infty$ and using $c_0>0$ with nonnegative terms gives (ii) and (iii). □

Lemma 3.10. $\lim_{k\to\infty}J_k^{\mathrm{cvx}}=0$.

Proof. If $I^{\mathrm{cvx}}=\emptyset$ then $J_k^{\mathrm{cvx}}=0$ for all k. Assume $I^{\mathrm{cvx}}\neq\emptyset$, fix $x^\star\in\Omega$, and put $B:=\overline{\mathrm{conv}}\{x^k:k\geq0\}$, compact by Lemma 3.9 and contained in $K\subset U$. By (A2) and compactness,

$$L:=\max_{i\in I^{\mathrm{cvx}}}\sup_{x\in B}\|\nabla F_i(x)\|<\infty,\qquad V:=\max_{i\in I}\sup_{x\in B}|F_i(x)-b_i|<\infty.$$

If $V = 0$ then all $v_i(x^k) = 0$ and $J_k^{\mathrm{cvx}} = 0$; assume $V > 0$. If $L = 0$ then each F_i, $i \in I^{\mathrm{cvx}}$, is constant on B, so $g_k^{\mathrm{cvx}} = 0$ and, by Remark 3, $0 = \langle g_k^{\mathrm{cvx}}, x^k - x^\star \rangle \geq 2\Phi_k^{\mathrm{cvx}} \geq (J_k^{\mathrm{cvx}})^2$, giving $J_k^{\mathrm{cvx}} = 0$; assume $L > 0$. Set $m^{\mathrm{cvx}} := |I^{\mathrm{cvx}}| \geq 1$.

Consider a step with $J_k^{\mathrm{cvx}} > 0$ and pick $i^\star \in I^{\mathrm{cvx}}$ with $v_{i^\star}(x^k) = J_k^{\mathrm{cvx}}$. Since $v_{i^\star}(x^k) \geq J_k^{\mathrm{cvx}} - \delta$, we have $i^\star \in K_k^{\mathrm{cvx}}$, whence

$$\Phi_k^{\mathrm{cvx}} \geq \tfrac{1}{2} v_{i^\star}(x^k)^2 = \tfrac{1}{2}(J_k^{\mathrm{cvx}})^2 > 0. \tag{25}$$

Remark 3 gives $\langle g_k^{\mathrm{cvx}}, x^k - x^\star \rangle \geq 2\Phi_k^{\mathrm{cvx}} > 0$, so $g_k^{\mathrm{cvx}} \neq 0$; with Lemma 3.6, $\|g_k^{\mathrm{cvx}}\|^2 \leq 2m^{\mathrm{cvx}} L^2 \Phi_k^{\mathrm{cvx}}$, and with (25),

$$T_k = \frac{(2\Phi_k^{\mathrm{cvx}})^2}{\|g_k^{\mathrm{cvx}}\|^2} \geq \frac{2\Phi_k^{\mathrm{cvx}}}{m^{\mathrm{cvx}} L^2} \geq \frac{(J_k^{\mathrm{cvx}})^2}{m^{\mathrm{cvx}} L^2}. \tag{26}$$

For a lower bound on μ_k: for all $i \in I$ and $x^k \in B$, $v_i(x^k) \leq |F_i(x^k) - b_i| \leq V$, so $J_k^{\mathrm{cvx}} \leq V$ and $J_k^{\mathrm{ps}} \leq V$; when $J_k^{\mathrm{cvx}} > 0$,

$$\mu_k = \frac{J_k^{\mathrm{cvx}}}{J_k^{\mathrm{cvx}} + J_k^{\mathrm{ps}}} \geq \frac{J_k^{\mathrm{cvx}}}{2V}. \tag{27}$$

Combining (26) and (27), for $J_k^{\mathrm{cvx}} > 0$, $\mu_k T_k \geq (J_k^{\mathrm{cvx}})^3/(2V m^{\mathrm{cvx}} L^2)$; this also holds (as $0 \geq 0$) when $J_k^{\mathrm{cvx}} = 0$. By Lemma 3.9(ii),

$$\sum_{k=0}^{\infty} (J_k^{\mathrm{cvx}})^3 \leq 2V m^{\mathrm{cvx}} L^2 \sum_{k=0}^{\infty} \mu_k T_k < \infty, \tag{28}$$

so $(J_k^{\mathrm{cvx}})^3 \to 0$ and thus $J_k^{\mathrm{cvx}} \to 0$. □

Proposition 3.11.

$$\sum_{k=0}^{\infty} \left\| x^{k+1} - x^k \right\|^2 < \infty;$$

in particular, $\|x^{k+1} - x^k\| \to 0$.

Proof. Since $x^k \in K$ and $x^{k+1} = P_K(y^k)$, Lemma 3.2 with $y = y^k$, $z = x^k$ gives $\|x^{k+1} - x^k\| \leq \|y^k - x^k\| = \|d_k\|$. By (24) and $\mu_k^2 \leq \mu_k$, $\|d_k\|^2 \leq (1+\zeta)\rho_{\mathrm{cvx}}^2 \mu_k T_k + C_0 \gamma_k^2$. Summing and using Lemma 3.9(ii) and (A3)(iii), both series are finite. □

3.8. Strong convergence to a feasible point

We now prove that the whole sequence converges strongly to a feasible point. The proof has three stages. Stage 1 uses Lemma 3.10 to show that every cluster point satisfies the convex constraints. Stage 2 shows, by contradiction and pseudoconvexity, that at least one cluster point is feasible. Stage 3 uses the convergence of $\|x^k - \bar{x}\|$ to show that the whole sequence converges to that point.

Theorem 3.12. *Under Assumption 1, the sequence $\{x^k\}$ generated by Algorithm 1 converges strongly to a point $x^\infty \in \Omega$.*

Proof. By Lemma 3.9, $\{x^k\}$ is bounded; let $C := \{\bar{x} \in H : \exists\, k_l \to \infty,\ x^{k_l} \to \bar{x}\}$ be its (nonempty, compact) cluster set. Since K is closed and $x^k \in K$, $C \subseteq K$.

Stage 1 (convex constraints). Fix $\bar{x} \in C$ and $x^{k_l} \to \bar{x}$. By (A2) each F_i is continuous, so $v_i(x^{k_l}) \to v_i(\bar{x})$; for $i \in I^{\mathrm{cvx}}$, $v_i(x^{k_l}) \le J_{k_l}^{\mathrm{cvx}} \to 0$ by Lemma 3.10, so by squeezing $v_i(\bar{x}) = 0$, i.e. $F_i(\bar{x}) \le b_i$. Hence

$$C \subseteq \bigcap_{i \in I^{\mathrm{cvx}}} \{x \in K : F_i(x) \le b_i\}. \tag{29}$$

If $I^{\mathrm{ps}} = \emptyset$, then (29) gives $C \subseteq \Omega$, so $C \cap \Omega = C \neq \emptyset$. Assume henceforth $I^{\mathrm{ps}} \neq \emptyset$.

Stage 2 (a feasible cluster point). Suppose, for contradiction, that $C \cap \Omega = \emptyset$. By (29) and the partition $I = I^{\mathrm{cvx}} \cup I^{\mathrm{ps}}$, for each $\bar{x} \in C$ there is $j \in I^{\mathrm{ps}}$ with $F_j(\bar{x}) > b_j$, i.e. $J^{\mathrm{ps}}(\bar{x}) > 0$. As a maximum of finitely many continuous functions, J^{ps} is continuous and positive on the compact set C, so $\alpha := \min_{x \in C} J^{\mathrm{ps}}(x) > 0$.

Claim 1 (uniform lower bound for J_k^{ps}). There exist $\tau_0 > 0$ and k_0 with $J_k^{\mathrm{ps}} \ge \tau_0$ for all $k \ge k_0$. Indeed, put $\tau_0 := \alpha/2$. If the claim failed there would be $k_l \to \infty$ with $J_{k_l}^{\mathrm{ps}} < \tau_0$; extracting (same notation) $x^{k_l} \to \tilde{x} \in C$, continuity gives $J^{\mathrm{ps}}(\tilde{x}) \le \tau_0 < \alpha$, contradicting the definition of α.

Claim 2 ($\mu_k \to 0$). For $k \ge k_0$, $\mu_k = J_k^{\mathrm{cvx}}/(J_k^{\mathrm{cvx}} + J_k^{\mathrm{ps}}) \le J_k^{\mathrm{cvx}}/\tau_0 \to 0$ by Lemma 3.10. Hence there is $k_1 \ge k_0$ with $1 - \mu_k \ge 1/2$ for $k \ge k_1$.

Claim 3 (uniform lower bound for $R_k^{x^\star}$). Fix $x^\star \in \Omega$. There exist $\xi_0 > 0$ and $k_2 \ge k_1$ with $R_k^{x^\star} \ge \xi_0$ for all $k \ge k_2$. Suppose not: there is $k_l \to \infty$ with $R_{k_l}^{x^\star} \to 0$. As $\{x^{k_l}\}$ is bounded, extract $x^{k_l} \to \bar{x} \in C$; since I^{ps} has finitely many subsets, extract further so that $K_{k_l}^{\mathrm{ps}} \equiv K^\star$ is constant. By assumption $J^{\mathrm{ps}}(\bar{x}) > 0$; choose $j^\star \in \arg\max_{j \in I^{\mathrm{ps}}} v_j(\bar{x})$, so $v_{j^\star}(\bar{x}) = J^{\mathrm{ps}}(\bar{x}) > 0$. By continuity $v_{j^\star}(x^{k_l}) \to v_{j^\star}(\bar{x})$ and $J_{k_l}^{\mathrm{ps}} \to J^{\mathrm{ps}}(\bar{x})$, so $J_{k_l}^{\mathrm{ps}} - v_{j^\star}(x^{k_l}) \to 0$; with $\delta > 0$ from (A3)(i), for large l we get $v_{j^\star}(x^{k_l}) \ge J_{k_l}^{\mathrm{ps}} - \delta$, hence $j^\star \in K_{k_l}^{\mathrm{ps}} = K^\star$. Each summand of $R_{k_l}^{x^\star} = \sum_{j \in K^\star} v_j(x^{k_l}) \langle \nabla F_j(x^{k_l}),\, x^{k_l} - x^\star \rangle$ is nonnegative by Lemma 3.5, so keeping only $j^\star$,

$$R_{k_l}^{x^\star} \ge v_{j^\star}(x^{k_l}) \left\langle \nabla F_{j^\star}(x^{k_l}),\, x^{k_l} - x^\star \right\rangle.$$

Letting $l \to \infty$ and using continuity of $v_{j^\star}$ and $\nabla F_{j^\star}$,

$$\liminf_{l \to \infty} R_{k_l}^{x^\star} \ge v_{j^\star}(\bar{x}) \langle \nabla F_{j^\star}(\bar{x}),\, \bar{x} - x^\star \rangle =: \Lambda.$$

Since $j^\star \in I^{\mathrm{ps}}$ and $x^\star \in \Omega$, $F_{j^\star}(x^\star) \le b_{j^\star} < F_{j^\star}(\bar{x})$; definition (5) with $x = \bar{x}$, $y = x^\star$ gives $\langle \nabla F_{j^\star}(\bar{x}),\, \bar{x} - x^\star \rangle > 0$, and with $v_{j^\star}(\bar{x}) > 0$ we get $\Lambda > 0$, contradicting $R_{k_l}^{x^\star} \to 0$. This proves Claim 3.

Reaching the contradiction. By Lemma 3.7 on $B := \overline{\mathrm{conv}}\{x^k\}$, $\|g_k^{\mathrm{ps}}\| \le |I^{\mathrm{ps}}|\, V_B L'_B$ with $L'_B := \max_{j \in I^{\mathrm{ps}}} \sup_B \|\nabla F_j\| < \infty$. Put $G' := \max\{1, |I^{\mathrm{ps}}|\, V_B L'_B\}$; then $\max\{1, \|g_k^{\mathrm{ps}}\|\} \le G'$ for all k, so $\eta_k^{\mathrm{ps}} = \gamma_k / \max\{1, \|g_k^{\mathrm{ps}}\|\} \ge \gamma_k / G'$. For $k \ge k_2$, combining with $1 - \mu_k \ge 1/2$ and $R_k^{x^\star} \ge \xi_0$,

$$\sum_{k=0}^{\infty} (1 - \mu_k) \eta_k^{\mathrm{ps}} R_k^{x^\star} \ge \sum_{k \ge k_2} \frac{1}{2} \cdot \frac{\gamma_k}{G'} \cdot \xi_0 = \frac{\xi_0}{2G'} \sum_{k \ge k_2} \gamma_k = +\infty$$

by (A3)(iii), contradicting Lemma 3.9(iii). Hence $C \cap \Omega \neq \emptyset$.

Stage 3 (whole-sequence convergence). Fix $\bar{x} \in C \cap \Omega$. Lemma 3.9(i), applied to $x^\star = \bar{x}$, ensures that $\lim_{k\to\infty} \left\|x^k - \bar{x}\right\|^2$ exists. Since $\bar{x} \in C$, a subsequence x^{k_l} converges to $\bar{x}$. Therefore

$$\lim_{k\to\infty} \left\|x^k - \bar{x}\right\|^2 = \lim_{l\to\infty} \left\|x^{k_l} - \bar{x}\right\|^2 = 0.$$

Hence the whole sequence converges to $\bar{x}$, and we set $x^\infty := \bar{x} \in \Omega$. The argument does not require Ω to be a singleton: one feasible cluster point and the convergence of its distance suffice. □

Remark 4. The parameter $\delta > 0$ is not necessary for convergence. When $\delta = 0$ one argues by subsequence: along $x^{k_l} \to \bar{x} \in C$, choose $j_l \in \arg\max_{j \in I^{\mathrm{ps}}} v_j(x^{k_l})$; since I^{ps} is finite, extract $j_l \equiv j^\star$, and $v_{j^\star}(x^{k_l}) = J^{\mathrm{ps}}(x^{k_l}) \to J^{\mathrm{ps}}(\bar{x}) > 0$, so $j^\star$ attains the maximal violation at $\bar{x}$ and the pseudoconvexity argument goes through. The role of $\delta > 0$ is to simplify: a maximizer at $\bar{x}$ automatically lies in $K^{\mathrm{ps}}_{k_l}$ for large l, avoiding a 'finitely many indices' extraction. Larger δ enlarges K^{cvx}_k and K^{ps}_k, which may improve numerical stability when several $v_i(x^k)$ are nearly equal.

Remark 5. From (28), $\sum_k (J^{\mathrm{cvx}}_k)^3 < \infty$, whence $\min_{0\le k<N} J^{\mathrm{cvx}}_k = O(N^{-1/3})$. This bounds the *best* convex-track violation over the first N iterates. It is not a last-iterate bound, a rate for the full residual $J(x^k)$, or a rate for the distance to a feasible point. For these latter quantities, under (A1)–(A3) alone we assert only qualitative convergence.

3.9. Degenerate constraint structures

When the constraint structure degenerates (only a convex track or only a pseudoconvex track), DTAP and its analysis simplify, yielding two corollaries of Theorem 3.12.

Corollary 3.13. *If $I^{\mathrm{ps}} = \emptyset$, then $J^{\mathrm{ps}}_k \equiv 0$, $g^{\mathrm{ps}}_k \equiv 0$, and $\mu_k \equiv 1$ whenever $J^{\mathrm{cvx}}_k > 0$, so Algorithm 1 reduces to $x^{k+1} = P_K(x^k - \eta^{\mathrm{cvx}}_k g^{\mathrm{cvx}}_k)$. The sequence converges strongly to a point $x^\infty \in \Omega = \{x \in K : F_i(x) \le b_i \text{ for all } i \in I^{\mathrm{cvx}}\}$.*

Proof. Here $d_k = \eta^{\mathrm{cvx}}_k g^{\mathrm{cvx}}_k$, and (21) directly gives $\left\|x^{k+1} - x^\star\right\|^2 \le \left\|x^k - x^\star\right\|^2 - 2\eta^{\mathrm{cvx}}_k \left\langle g^{\mathrm{cvx}}_k, x^k - x^\star\right\rangle + (\eta^{\mathrm{cvx}}_k)^2 \left\|g^{\mathrm{cvx}}_k\right\|^2$. If $\|g^{\mathrm{cvx}}_k\| = 0$ then $\eta^{\mathrm{cvx}}_k = 0$, $T_k = 0$, and the bound is $\left\|x^{k+1} - x^\star\right\|^2 \le \left\|x^k - x^\star\right\|^2$. Otherwise Lemma 3.4 gives $\left\langle g^{\mathrm{cvx}}_k, x^k - x^\star\right\rangle \ge 2\Phi^{\mathrm{cvx}}_k$, and substituting $\eta^{\mathrm{cvx}}_k = 2\rho_{\mathrm{cvx}}\Phi^{\mathrm{cvx}}_k / \left\|g^{\mathrm{cvx}}_k\right\|^2$ together with $(\eta^{\mathrm{cvx}}_k)^2 \left\|g^{\mathrm{cvx}}_k\right\|^2 = \eta^{\mathrm{cvx}}_k \cdot 2\rho_{\mathrm{cvx}}\Phi^{\mathrm{cvx}}_k$ yields, after simplification, $2\eta^{\mathrm{cvx}}_k \left\langle g^{\mathrm{cvx}}_k, x^k - x^\star\right\rangle - (\eta^{\mathrm{cvx}}_k)^2 \left\|g^{\mathrm{cvx}}_k\right\|^2 \ge \rho_{\mathrm{cvx}}(2 - \rho_{\mathrm{cvx}})(2\Phi^{\mathrm{cvx}}_k)^2 / \left\|g^{\mathrm{cvx}}_k\right\|^2 = 2c_0 T_k$. Hence

$$\left\|x^{k+1} - x^\star\right\|^2 \le \left\|x^k - x^\star\right\|^2 - 2c_0 T_k, \tag{30}$$

valid for all k. Thus $\{\left\|x^k - x^\star\right\|^2\}$ is nonincreasing and convergent, and summing gives $\sum_k T_k < \infty$. As in Lemma 3.10, when $J^{\mathrm{cvx}}_k > 0$ one has $T_k \ge (J^{\mathrm{cvx}}_k)^2/(m^{\mathrm{cvx}}L^2)$ (note the exponent 2 here, versus 3 in (28), since no μ_k factor is present), so $\sum_k (J^{\mathrm{cvx}}_k)^2 < \infty$ and $J^{\mathrm{cvx}}_k \to 0$. Finally, $I^{\mathrm{ps}} = \emptyset$ makes Stage 2 of Theorem 3.12 vacuous (indeed $C \subseteq \Omega$ already from (29)), and Stage 3 gives $x^k \to x^\infty \in \Omega$. □

Corollary 3.14. *If $I^{\mathrm{cvx}} = \emptyset$, then $J_k^{\mathrm{cvx}} \equiv 0$, $g_k^{\mathrm{cvx}} \equiv 0$, and $\mu_k \equiv 0$ whenever $J_k^{\mathrm{ps}} > 0$, so Algorithm 1 reduces to $x^{k+1} = P_K(x^k - \eta_k^{\mathrm{ps}} g_k^{\mathrm{ps}})$ with $\eta_k^{\mathrm{ps}} = \gamma_k / \max\{1, \|g_k^{\mathrm{ps}}\|\}$. The sequence converges strongly to a point $x^\infty \in \Omega$.*

Proof. Now $d_k = \eta_k^{\mathrm{ps}} g_k^{\mathrm{ps}}$ with $\|d_k\|^2 \le \gamma_k^2$, so (21) gives $\|x^{k+1} - x^\star\|^2 \le \|x^k - x^\star\|^2 - 2\eta_k^{\mathrm{ps}} R_k^{x^\star} + \gamma_k^2$, which is (20) in this degenerate case ($T_k = 0$, $1 - \mu_k = 1$). Since $R_k^{x^\star} \ge 0$ (Lemma 3.5), $a_{k+1} \le a_k + \gamma_k^2$, and Lemma 3.3 gives convergence of $\{\|x^k - x^\star\|^2\}$ and boundedness; summation gives $\sum_k \eta_k^{\mathrm{ps}} R_k^{x^\star} < \infty$. Stage 1 of Theorem 3.12 gives $C \subseteq K$ (the convex part being vacuous); Stage 2 with $\mu_k = 0$ (so $1 - \mu_k = 1$) produces the same contradiction $\sum \eta_k^{\mathrm{ps}} R_k^{x^\star} = +\infty$, so $C \cap \Omega \neq \emptyset$; Stage 3 gives $x^k \to x^\infty \in \Omega$. □

4. Numerical experiments

We separate predictive performance from the numerical solution of the feasibility system. The classification comparison uses eight datasets. The solver diagnostic additionally includes Movement Libras and controlled nonconvex pseudoconvex instances.

4.1. Feature-selection model

We apply the normal-box NSFP to feature selection by combining criteria from the fractional-programming model of Wang et al. [15] and the fuzzy-rough model AFRS-FCV [16]. Let $\mathcal{X} = \{x_1, \dots, x_n\}$ be the training samples, $A = \{a_1, \dots, a_p\}$ the features, $X \in \mathbb{R}^{n \times p}$ the data matrix, and $y \in \{1, \dots, r\}^n$ the class labels. Here n is the number of samples; subscripts index features and superscripts index iterations. The feature weights belong to the compact convex simplex $K = \Delta_{p-1} = \{w \in \mathbb{R}^p : w \ge 0,\ \mathbf{1}_p^\top w = 1\}$. All supervised quantities below are estimated from the training data.

4.1.1. Fractional criterion and N-Fscore

The supervised similarity matrix $S \in \mathbb{R}^{p \times p}$ has entries

$$s_{ij} = \max\left\{0, \frac{I(Z_i; Z_j; C)}{H(Z_i) + H(Z_j)}\right\}, \qquad i, j = 1, \dots, p, \tag{31}$$

where Z_i is the discretized feature a_i, C is the class variable, and $I(Z_i; Z_j; C) = I(Z_i; C) + I(Z_j; C) - I((Z_i, Z_j); C)$ is their interaction information. Entropy and mutual information are estimated from equal-width histograms with ten bins per feature [2]. We set $s_{ij} = 0$ if the denominator is zero. The redundancy matrix is

$$Q_R = S + \delta_Q I_p, \qquad \delta_Q \ge -\min\{0, \lambda_{\min}(S)\}, \tag{32}$$

so $Q_R \succeq 0$. The implementation uses $\delta_Q = \max\{0, -\lambda_{\min}(S)\} + 10^{-10}$. Let n_t, $\bar{a}_{kt}$, and σ_{kt}^2 be the class size, mean, and population variance of feature a_k in class t, and let $\bar{a}_k$ be its overall mean. The Fisher scores are

$$\rho_k = \frac{\sum_{t=1}^r n_t (\bar{a}_{kt} - \bar{a}_k)^2}{\sum_{t=1}^r n_t \sigma_{kt}^2 + \varepsilon_\rho}, \qquad \varepsilon_\rho = 10^{-12}. \tag{33}$$

These scores are nonnegative. The denominator regularization prevents division by zero but does not ensure $\rho_k > 0$, since the numerator can vanish.

The fractional program underlying N-Fscore is

$$\min_{w \in K} \frac{w^\top Q_R w}{\rho^\top w}, \tag{34}$$

assuming the denominator is positive on K. The numerator measures redundancy and the denominator relevance. Our N-Fscore baseline uses the projection-neural-network stage of [15], with the Fisher scores in (33). It then applies the common selection rule (44), instead of the source method's k-winners-take-all stage. Its dynamics and explicit-Euler implementation are given in Section 4.2.1.

4.1.2. Fuzzy-rough criterion and AFRSFCV

For each class, write $D_t = \{x_i : y_i = t\}$ and $D_t(x_j) = \mathbf{1}_{\{y_j = t\}}$. AFRSFCV incorporates class variance into a fuzzy relation. With $a_k(x_i) = X_{ik}$, the implemented relation is

$$R_k(x_i, x_j) = \exp\left(-\frac{[a_k(x_i) - a_k(x_j)]^2}{2\widetilde{\sigma}^2_{k,y_i}}\right), \qquad \widetilde{\sigma}^2_{kt} = \max\{\sigma^2_{kt}, 10^{-10}\}. \tag{35}$$

The variance floor makes the formula defined when a feature is constant within a class. The fuzzy lower approximation is

$$\underline{R_k}(D_t)(x_i) = \min_{x_j \in \mathcal{X}} \max\{1 - R_k(x_i, x_j), D_t(x_j)\}. \tag{36}$$

Its values form the matrix $\underline{R_k}(D) \in [0,1]^{n \times r}$. Let $Y_{it} = \mathbf{1}_{\{y_i = t\}}$ be the label matrix and define the decision matrix by

$$P_{ik} = \mathbf{1}_{\{\arg\max_{1 \le t \le r} \underline{R_k}(D_t)(x_i) = y_i\}}. \tag{37}$$

Ties are resolved by the smallest class index. Thus $P_{ik} = 1$ precisely when feature a_k predicts sample x_i correctly under this rule.

The AFRSFCV baseline [16] minimizes

$$\begin{aligned} \min_{\tau \in \mathbb{R}^p} \; L(\tau) = \frac{1}{2} \left\| Y - \sum_{k=1}^{p} w_k(\tau) \underline{R_k}(D) \right\|_F^2 \\ + \frac{\lambda_1}{2} \|\mathbf{1}_n - P w(\tau)\|^2 + \frac{\lambda_2}{2} \|w(\tau)\|^2, \end{aligned} \tag{38}$$

where $w_k(\tau) = e^{\tau_k} / \sum_{l=1}^{p} e^{\tau_l}$ ensures $w(\tau) \in K$. The three terms are the label-approximation error, the weight of misclassifying features per sample, and a Tikhonov term. The baseline uses gradient descent for a fixed iteration budget, followed by (44). The gradient and update are given in Section 4.2.1.

4.1.3. Feasibility and feature selection

We impose separate bounds on three criteria,

$$F_1(w) = \left\| Y - \sum_{k=1}^{p} w_k \underline{R_k}(D) \right\|_F^2, \tag{39}$$

$$F_2(w) = \|\mathbf{1}_n - Pw\|^2, \tag{40}$$

$$F_3(w) = \frac{w^\top Q_R w}{\rho^\top w}, \qquad U_\rho = \{w : \rho^\top w > 0\}. \tag{41}$$

Here F_1, F_2 are the first two terms of (38), without their coefficients, and F_3 is the objective in (34). For thresholds $b_i > 0$, the resulting problem is

$$\text{find } w^\star \in \Omega, \qquad \Omega = \{w \in K : F_i(w) \le b_i,\ i = 1, 2, 3\}. \tag{42}$$

This formulation asks for simultaneous acceptability of the three criteria. It omits the Tikhonov term and does not preserve either source optimization problem or its minimizers.

Proposition 4.1. *If $Q_R \succeq 0$ and $\min_k \rho_k > 0$, all three functions in (39)–(41) are smooth and convex on an open convex neighbourhood of K. Consequently, (42) is an all-convex normal-box NSFP. Hence (42) satisfies (A1) and (A2) of Assumption 1, with $U = U_\rho$, whenever $\Omega \neq \varnothing$.*

Proof. The first two functions are squared norms of affine maps. For $w \in U_\rho$ and any direction h, put $t = \rho^\top w$ and $a = \rho^\top h$. Direct differentiation gives

$$D^2 F_3(w)[h, h] = \frac{2}{t} \left(h - \frac{a}{t} w\right)^\top Q_R \left(h - \frac{a}{t} w\right) \ge 0. \tag{43}$$

Thus F_3 is convex and smooth on the open half-space U_ρ. Since $\rho^\top w \ge \min_k \rho_k > 0$ for $w \in K$, this half-space contains K. □

The experiments use $I^{\mathrm{cvx}} = \{1, 2\}$, $I^{\mathrm{ps}} = \{3\}$, which is admissible since convex functions are pseudoconvex; DTAP then uses only sign information for F_3, so no genuinely nonconvex constraint is tested. The all-convex assignment $I^{\mathrm{cvx}} = \{1, 2, 3\}$ is also admissible, and Theorem 3.12 applies to both. Proposition 4.1 needs $\min_k \rho_k > 0$; on all instances used here the smallest score is 4.77×10^{-5}. Hence the safeguard $\max\{\rho^\top w, 10^{-15}\}$ in the implementation is never active, and the implemented F_3 is (41).

The thresholds are calibrated on the training data: b_i is the q_i-th percentile of F_i over 3 000 Dirichlet points (seed 42) and the p vertices of K, where a fixed rule chooses q_i from the sample size, the number of features and classes, and the class imbalance (supplementary metadata). All solvers share these thresholds. Calibrating each b_i separately does not guarantee $\Omega \neq \varnothing$; Section 4.4 checks feasibility numerically. Prescribed thresholds would be used as given.

For every method, sort the computed weights as $w^\star_{(1)} \ge \cdots \ge w^\star_{(p)}$ and retain the

first s features, where

$$s = \min\left\{ s' \in \{1, \dots, p\} : \sum_{j=1}^{s'} w^{\star}_{(j)} \geq \theta \right\}, \qquad 0 < \theta < 1. \tag{44}$$

This cumulative-weight rule does not enforce a prescribed feature budget. Truncating the weights does not preserve the feasibility guarantee for the full vector.

Remark 6. With the precomputed matrices, evaluating $F_1, \nabla F_1$ costs $O(nrp)$, evaluating $F_2, \nabla F_2$ costs $O(np)$, and evaluating $F_3, \nabla F_3$ costs $O(p^2)$. Projection onto $K = \Delta_{p-1}$ by sorting costs $O(p \log p)$ operations [6]. A dense iteration therefore costs $O(nrp + p^2)$ operations and requires $O(nrp + p^2)$ storage, including Q_R. Preprocessing and classifier training are separate costs.

4.2. Experimental setup

The main classification study uses the eight UCI datasets [8] in Table 1. The DTAP runs reported here start at the uniform weight vector and use $\rho_{\mathrm{cvx}} = 1.98$, $\delta = 10^{-3}$, $\gamma_k = (k + 0.5)^{-0.55}$, tolerance 10^{-6}, and a budget of 1000 iterations. SG-const uses the step $\nu_k = 10^{-3}$ and SG-dyn the sampled modulus $\widehat{L}$ (Section 4.2.1); tuned moduli are considered only in Section 4.4. N-Fscore uses $\varepsilon_{\mathrm{PNN}} = 10^{-3}$ and 1000 Euler grid points on $[0, 0.2]$. AFRSFCV uses learning rate 10^{-3}, $\lambda_1 = 1$, $\lambda_2 = 0.5$, and 1000 steps. The shared cumulative threshold is $\theta = 0.9$.

Table 1. Datasets used for the main classification comparison. The additional Movement Libras instance is used only in the solver diagnostic.

Dataset	Samples n	Features p	Classes r	Balance
Glass	214	9	6	Imbalanced
Diabetes	768	8	2	Imbalanced
Segmentation	2 310	18	7	Balanced
Abalone	4 177	8	3	Balanced
Satimage	6 435	36	6	Imbalanced
Yeast	1 484	7	10	Imbalanced
Spambase	4 601	57	2	Imbalanced
Pendigits	10 992	16	10	Balanced

4.2.1. Baseline solvers

The following paragraphs specify the baseline solvers: N-Fscore and AFRSFCV for the models of Sections 4.1.1 and 4.1.2, and the subgradient methods SG-const and SG-dyn.

N-Fscore: the projection neural network To solve the fractional program (34), Wang et al. [15] proposed a neurodynamic method coupling a projection neural net-

work (PNN) with a k-winners-take-all network. The PNN is the differential system

$$\begin{aligned}\varepsilon_{\mathrm{PNN}}\frac{du}{dt} = & -\Pi g(u)+\tfrac{1}{p}\mathbf{1}_p \\ & -(I-\Pi)\big(u-g(u)+\nabla f((I-\Pi)g(u)+\tfrac{1}{p}\mathbf{1}_p)\big).\end{aligned} \tag{45}$$

with output $w = g(u)$, where $g(u) = \max\{0, u\}$ is the componentwise ReLU, $\Pi = \frac{1}{p}\mathbf{1}_p\mathbf{1}_p^\top$, ∇f is the gradient of the objective in (34), and $\varepsilon_{\mathrm{PNN}} > 0$ is a time constant. We integrate (45) by the explicit Euler method on a uniform grid of N_{grid} points in $[0, T]$, without a stopping rule, using the parameters of Section 4.2. The weight vector is the projection $P_K\big(g(u(T))\big)$ of the final network output onto K. For a uniform comparison we use only the PNN part to obtain $w^\star$; the k-WTA stage is replaced by the shared cumulative-threshold rule of Section 4.1.3.

AFRSFCV: gradient and update The softmax loss (38) is minimized over $\tau \in \mathbb{R}^p$ by gradient descent. Its gradient is

$$\nabla_\tau L(\tau) = \big(\mathrm{diag}(w(\tau)) - w(\tau)w(\tau)^\top\big)\nabla_w L(w(\tau)), \tag{46}$$

with $\nabla_w L = -\mathbf{M}^\top(\mathrm{vec}(Y) - \mathbf{M}w) + \lambda_1(P^\top Pw - P^\top\mathbf{1}_n) + \lambda_2 w$, where $\mathbf{M} \in \mathbb{R}^{nr\times p}$ has k-th column $\mathrm{vec}(\underline{R_k}(D))$. The update $\tau^{t+1} = \tau^t - \eta\,\nabla_\tau L(\tau^t)$ runs for a fixed budget of T steps; the final weight $w^\star = w(\tau^T) \in K$ feeds the shared selection rule of Section 4.1.3.

SG-const and SG-dyn These subgradient methods for quasiconvex feasibility [7] follow its most-violated-constraint strategy: with $\mathcal{I}_k = \arg\max_i v_i(w^k)$,

$$w^{k+1} = P_K(w^k - \nu_k\bar{h}_k), \qquad \bar{h}_k = \frac{1}{|\mathcal{I}_k|}\sum_{i\in\mathcal{I}_k}\frac{\nabla F_i(w^k)}{\|\nabla F_i(w^k)\|}.$$

SG-const uses $\nu_k = 10^{-3}$; SG-dyn uses the dynamic step $\nu_k = J(w^k)/L$ (Hölder exponent 1), with $L = \widehat{L}$, the largest gradient norm at 500 points sampled in K.

4.2.2. Evaluation protocol

Each dataset uses a single stratified 80/20 training-pool/test split (seed 42). Min–max scaling is fitted once on the full training pool, not separately within its selection folds, and is then applied to the test data. Stratified folds (seed 42) within the pool use $N_f = 10$, reduced to 7 on Glass and 4 on Yeast. The fold masks use $\theta = 0.9$ in (44); the fixed strict-majority rule retains features selected in at least $\lfloor N_f/2\rfloor + 1$ folds. We denote the voted feature set by $A^\star$. The final classifiers use the full pool with these voted features: RBF-SVM ($C = 1$, `gamma=scale`) and 5-NN, with the full feature set as a reference. Accuracy and macro one-vs-rest AUC are reported.

The voted masks are computed once and then reused: the evaluation stage trains only the downstream classifiers on these fixed masks, with SVM probability-calibration seed 42; identical masks share the same predictions. The test data are not used to select features or to choose any parameter or vote threshold. The training/test split is fixed before the selections are computed, and the rule that chooses the threshold levels uses

only the sample size, the number of features, the number of classes and the class sizes of the whole dataset, not any feature values. Because a single split is used, this is an exploratory evaluation, not an end-to-end or nested-validation study. For the fold-selection matrix $Z \in \{0,1\}^{N_f \times p}$, the Nogueira index [13] is computed as

$$\Psi = 1 - \frac{p^{-1}\sum_{j=1}^{p} s_j^2}{\bar{q}(1-\bar{q})}, \quad s_j^2 = \frac{1}{N_f - 1}\sum_{b=1}^{N_f}(Z_{bj} - \bar{Z}_j)^2, \quad \bar{q} = \frac{1}{N_f p}\sum_{b,j} Z_{bj}.$$

Here $\bar{Z}_j = N_f^{-1}\sum_b Z_{bj}$, and s_j^2 is the unbiased variance over the N_f folds. Since the training folds overlap, stability is reported descriptively. For each classifier metric, we use a two-sided signed-rank test across the eight datasets, omitting zero differences, assigning midranks to ties, and enumerating all signs of the nonzero differences; all-zero pairs receive the p-value 1. Holm adjustment covers the four DTAP–baseline comparisons separately for each metric. The Friedman test compares the five selection methods.

4.3. Classification results

Tables 2–4 report selected-set size, stability, and predictive performance. DTAP retains 84.2% of the features on average, including 33/36 on Satimage and 49/57 on Spambase. Its mean stability is 0.786, compared with N-Fscore's 0.822. Thus DTAP reduces the number of features moderately; it does not select sparser or more stable sets than the base models. AFRSFCV selects much smaller sets on several datasets, with lower mean accuracy under the fixed configuration.

DTAP's mean accuracy is within 0.44 percentage points of the full-feature reference for both classifiers. It exceeds N-Fscore and AFRSFCV by 3.12 and 4.83 points on SVM accuracy, while retaining more features. This is a size–prediction tradeoff, not a matched-budget comparison. Both SG variants have exactly the same voted masks as DTAP on seven datasets; only Spambase differs (48 versus 49 features). SG-dyn used the full 1000-iteration budget in 6/7 selection folds on Glass, 7/10 on Diabetes and 10/10 on Abalone, and SG-const in 1/10 on Abalone (DTAP in none); the voted weights of both SG variants are the best iterates found within the budget. The SG results on Spambase, the only dataset where they differ from DTAP, are given below Table 3; Table 4 includes all five methods. Fold-wise stability can differ even when the final voted mask is identical.

Table 5 reports the signed-rank tests. None of the pairwise comparisons rejects at 0.05 after Holm adjustment; the SVM comparisons with the two base models have adjusted p-values between 0.0625 and 0.0938. The asymptotic Friedman tests reject equality globally, but this does not establish a particular pairwise superiority. All DTAP–SG pairwise p-values equal 1; this reflects ties and limited evidence, not a statistical equivalence claim. These small-sample comparisons concern eight heterogeneous datasets and do not measure variation under repeated train/test splits.

4.4. Constraint activity and a tuned subgradient comparison

As noted in Section 4.1.3, the calibrated thresholds need not yield $\Omega \neq \varnothing$. For each full-data instance below, we keep the evaluated point with the largest slack $\min_i(b_i - F_i(w))$ over all runs. Re-evaluated on matrices rebuilt from the data, each of these points

Table 2. Selected count $|A^\star|$ / Nogueira stability Ψ; p is the original dimension. All five selection methods are shown.

Dataset	p	N-Fscore	AFRSFCV	DTAP	SG-const	SG-dyn
Glass	9	7/0.816	7/0.609	7/0.633	7/0.633	7/0.633
Diabetes	8	7/0.290	3/0.810	7/0.707	7/0.707	7/0.707
Segmentation	18	8/0.885	2/1.000	16/0.800	16/0.888	16/0.888
Abalone	8	4/0.950	2/1.000	5/1.000	5/1.000	5/0.870
Satimage	36	10/0.871	3/0.685	33/0.563	33/0.563	33/0.563
Yeast	7	6/1.000	5/1.000	6/0.757	6/0.757	6/0.757
Spambase	57	26/0.910	4/0.885	49/0.832	48/0.851	48/0.851
Pendigits	16	13/0.854	12/0.063	15/1.000	15/1.000	15/1.000

Table 3. Per-dataset accuracy (%) / macro one-vs-rest AUC on the single holdout. Equal voted masks reuse the same predictions; the SG exceptions are stated below.

Dataset	Classifier	Full	N-Fscore	AFRSFCV	DTAP
Glass	SVM	76.74/0.9222	69.77/0.9073	72.09/0.9174	72.09/0.9174
	5-NN	72.09/0.9097	74.42/0.8933	69.77/0.9371	69.77/0.9371
Diabetes	SVM	74.68/0.8089	71.43/0.7957	65.58/0.6617	74.68/0.8098
	5-NN	71.43/0.7356	70.78/0.7154	67.53/0.7145	73.38/0.7766
Segmentation	SVM	93.29/0.9936	80.52/0.9714	78.79/0.9668	93.29/0.9940
	5-NN	94.37/0.9880	87.01/0.9801	86.58/0.9646	94.59/0.9897
Abalone	SVM	61.60/0.8102	61.36/0.8065	62.56/0.8114	62.68/0.8123
	5-NN	59.81/0.7742	61.00/0.7605	58.73/0.7752	60.29/0.7695
Satimage	SVM	89.12/0.9796	87.10/0.9756	80.42/0.9611	89.12/0.9797
	5-NN	91.14/0.9792	87.80/0.9705	80.96/0.9306	91.22/0.9766
Yeast	SVM	58.25/0.8561	58.92/0.8241	57.91/0.8207	58.59/0.8426
	5-NN	51.52/0.7667	51.52/0.7763	51.52/0.7681	51.52/0.7585
Spambase	SVM	92.07/0.9662	88.38/0.9392	87.62/0.9155	91.97/0.9616
	5-NN	90.34/0.9443	85.78/0.9214	88.49/0.9206	89.25/0.9437
Pendigits	SVM	99.09/0.9995	98.95/0.9996	97.77/0.9996	98.95/0.9995
	5-NN	98.95/0.9982	98.64/0.9982	97.73/0.9965	99.05/0.9987

Both SG variants equal DTAP exactly on the other seven datasets. On Spambase, both give 91.86/0.9628 (SVM) and 89.14/0.9441 (5-NN).

Table 4. Unweighted mean classification performance over the eight classification datasets. All five selection methods and the full-feature reference are included.

Method	SVM accuracy	5-NN accuracy	SVM AUC	5-NN AUC
Full	80.61	78.71	0.9170	0.8870
N-Fscore	77.05	77.12	0.9024	0.8770
AFRSFCV	75.34	75.16	0.8818	0.8759
DTAP	80.17	78.63	0.9146	0.8938
SG-const	80.16	78.62	0.9148	0.8939
SG-dyn	80.16	78.62	0.9148	0.8939

Table 5. DTAP versus each baseline: raw / Holm-adjusted signed-rank p-values. Each metric has four comparisons over eight datasets. The last column gives the tie-corrected Friedman asymptotic p-value for all five methods.

Metric	N-Fscore	AFRSFCV	SG-const	SG-dyn	Friedman
Accuracy, SVM	0.0312/0.0938	0.0156/0.0625	1.0000/1.0000	1.0000/1.0000	0.0005
Accuracy, 5-NN	0.3750/1.0000	0.0312/0.1250	1.0000/1.0000	1.0000/1.0000	0.0057
AUC, SVM	0.0156/0.0625	0.0312/0.0938	1.0000/1.0000	1.0000/1.0000	0.0055
AUC, 5-NN	0.0781/0.3125	0.1562/0.4688	1.0000/1.0000	1.0000/1.0000	0.0435

Table 6. Optimization evaluations from the uniform simplex start. AC denotes the all-convex comparator; sampled and tuned refer to SG-dyn. † means the tolerance was not reached within the budget. The last column gives the DTAP third-constraint violation count divided by its number of evaluated points.

Dataset	DTAP	AC	SG-const	Sampled	Tuned	F_3 activity
Glass	89	89	520	1000†	100	0/89
Diabetes	14	14	350	1000†	51	0/14
Segmentation	3	3	124	170	3	0/3
Abalone	48	48	638	1000†	3	1/48
Satimage	2	2	60	102	2	0/2
Yeast	4	4	324	305	3	0/4
Spambase	5	5	93	295	56	0/5
Pendigits	2	2	49	10	31	0/2
Movement Libras	637	637	62	1000†	144	0/637

satisfies all three inequalities strictly; the smallest slack is 4.33×10^{-4}. This is a numerical check, not an exact certificate of feasibility.

We count the iterates of a run, including the last (its evaluated points), at which $F_3(w^k) - b_3 > 10^{-6}$. This measures violation along the trajectory, rather than whether the constraint is binding at its limit. Table 6 shows that, from the standard start, F_3 is violated at no evaluated point on seven of the eight classification datasets, and at one of the 48 evaluated points on Abalone. Since F_3 is convex (Proposition 4.1), these feature-selection instances test a convex subclass and do not demonstrate a need for the nonconvex extension. We also report the all-convex comparator (AC), which assigns all three constraints to the convex track.

Protocol and tuning cost. These are optimization-only tests on full-data feasibility instances. They use $J \le 10^{-6}$, at most 1 000 evaluated points, and identical starting points across methods. SG-const and SG-dyn are defined in Section 4.2.1. Besides SG-dyn with the sampled modulus $\widehat{L}$, a tuned variant selects $L = c\widehat{L}$ from the manually specified grid $c \in \{10^{-3}, 10^{-2.5}, \dots, 10^{1}\}$ using three random starting points (seeds 1101–1103). Selection first maximizes the number of successful runs, then minimizes total evaluations, then total best residual. Evaluation uses the standard start and three different random starts (seeds 2101–2103). The feature starts are Dirichlet samples. Only starting points are held out: tuning and evaluation use the same full-data instance, and no classification score is used. Neither $\widehat{L}$ nor the tuned value is a certified global Hölder modulus.

Whenever $F_3(w^k) \le b_3$ along a run, the pseudoconvex track is empty, $\mu_k = 1$ and DTAP coincides step by step with the all-convex comparator; on the eight instances with zero F_3 activity, the identical DTAP and AC columns reflect this identity. On

Table 7. Nonconvex pseudoconvex tests, grouped by κ. Each entry is successes out of 12 runs, followed by the median number of evaluated points in parentheses, including capped runs. Each group contains all three dimensions and four starts. Tuned SG-const uses the step selected as described in the text.

κ	DTAP	Balanced	SG-const	Tuned SG-const	Sampled SG-dyn	Tuned SG-dyn
0.3	0 (1000)	0 (1000)	0 (1000)	12 (15)	8 (725.5)	12 (142.5)
1	12 (70)	12 (395.5)	0 (1000)	12 (15)	12 (301.5)	12 (44)
3	12 (4)	12 (4)	0 (1000)	12 (16)	12 (167)	12 (11)

Abalone, F_3 is violated only at the uniform start (1/48), where $J_0^{\mathrm{ps}} \approx 1.56 \times 10^{-3}$ is negligible against $J_0^{\mathrm{cvx}} \approx 366.3$, so $1 - \mu_0 \approx 4.27 \times 10^{-6}$; the values of J along the two trajectories satisfy $|J_{\mathrm{DTAP}} - J_{\mathrm{AC}}| \le 3.18 \times 10^{-5}\, J_{\mathrm{AC}}$ at every evaluated point, and both runs stop after 48 evaluated points. The columns are therefore not a comparison of the two assignments, whose success counts differ only on the random starts (26/36 versus 36/36).

Tuning raises SG-dyn's standard-start success count from 5/9 to 9/9. It can outperform DTAP, for example on Abalone, and SG-const is faster on Movement Libras. Across all four starts on all nine feature instances, the success counts are 26/36 for DTAP, 36/36 for the all-convex comparator, 33/36 for SG-const, 15/36 for sampled SG-dyn, and 28/36 for tuned SG-dyn. A balanced DTAP variant also obtains 26/36; it assigns equal weights when both tracks are violated and full weight to the only violated track otherwise. Thus these tests do not establish a uniform advantage of the proportional weight or robustness to initialization.

The nine feature-instance searches required 179 744 optimization evaluations and 10.80 seconds of tuning solver time in total. These costs are additional to Table 6; DTAP was not tuned. Initial modulus sampling, preprocessing and search overhead are excluded from this solver-time total. The supplementary records give the cost of each search separately.

A genuinely nonconvex pseudoconvex constraint. To exercise the pseudoconvex track directly, consider $K = [-2, 2]^d$ and

$$\begin{aligned} F_1(x) &= \|x + e_1\|^2 \le 1, \qquad F_2(x) = \|x - e_2\|^2 \le \tfrac{9}{4}, \\ F_3(x) &= \kappa \log(1 + \|x - e_1\|^2) \le \kappa \log 3. \end{aligned} \tag{47}$$

Here $d \in \{2, 10, 50\}$ and $\kappa \in \{0.3, 1, 3\}$ give nine configurations. The third function is pseudoconvex: it is an increasing differentiable transform, with positive derivative, of a convex quadratic. It is not convex, since its radial Hessian eigenvalue is $2\kappa(1-r^2)/(1+r^2)^2 < 0$ for $r = \|x - e_1\| > 1$. The exact witness $x = -e_1/10$ has positive slacks 19/100, 31/25, and $\kappa \log(300/221)$. The standard start is $1.5\mathbf{1}$; the three evaluation starts and three tuning starts are drawn uniformly from $[0.8, 1.8]^d$ with the seeds above (NumPy `default_rng`). For modulus sampling we use 500 uniform points in K (seed 42). SG-const uses the same step $\nu_k = 10^{-3}$ as on the feature instances. Tuned SG-const uses a constant step ν selected from the grid $\{10^{-3}, 10^{-2.5}, \ldots, 10^{1}\}$ with the tuning starts and selection rule of tuned SG-dyn; it selects $\nu = 10^{-0.5}$ in all nine configurations. The balanced variant is the one described above.

DTAP encounters a third-constraint violation in every one of these 36 runs. Its pooled activity is 12 835/12 877 evaluated points. Nevertheless, it reaches the tolerance in only 24/36 runs, compared with 32/36 for sampled SG-dyn, 36/36 for tuned SG-

Table 8. DTAP on (47) with one parameter changed at a time, $\gamma_k = a(k+0.5)^{-s}$. The baseline $\rho_{\mathrm{cvx}} = 1.98$, $\delta = 10^{-3}$, $a = 1$, $s = 0.55$ is the DTAP column of Table 7. Each entry is successes out of 12 runs (three dimensions, four starts), followed by the median number of evaluated points in parentheses, including capped runs.

Setting	$\kappa = 0.3$	$\kappa = 1$
Baseline	0 (1000)	12 (70)
$\rho_{\mathrm{cvx}} = 1$	4 (1000)	12 (223.5)
$\delta = 10^{-1}$	0 (1000)	12 (70)
$a = 10$	12 (77)	12 (6)
$a = 100$	12 (6)	12 (48)
$s = 0.75$	0 (1000)	12 (374)
$s = 1$	0 (1000)	0 (1000)

dyn and 36/36 for tuned SG-const (Table 7). Tuned SG-const reaches the tolerance in every run with a median of 15–16 evaluated points, fewer than any other method at $\kappa = 0.3$ and $\kappa = 1$; at $\kappa = 3$, DTAP and the balanced variant need a median of 4. These counts exclude the 147 331 evaluations of its tuning search.

With $\nu_k = 10^{-3}$, SG-const moves at most $999 \times 10^{-3} = 0.999$ from its start within the budget, whereas every start lies at distance at least 1.11 from Ω; its zero counts are therefore fixed by the step size and the budget. For $\kappa \le 1$ the convex track is satisfied from the second evaluated point on, so DTAP then follows the single-track iteration of Corollary 3.14; both tracks are violated simultaneously at only 32 of these 12 877 evaluated points.

All DTAP failures occur at $\kappa = 0.3$. Changing κ leaves the feasible set unchanged, but v_3 and ∇F_3 both scale with κ, so when $\|g_k^{\mathrm{ps}}\| \le 1$ the pseudoconvex step $(1 - \mu_k)\gamma_k g_k^{\mathrm{ps}}$ scales with $\gamma_k \kappa^2$. Table 8 changes one DTAP parameter at a time for $\kappa \in \{0.3, 1\}$. With $\gamma_k = a(k + 0.5)^{-0.55}$, $a = 10$ and $a = 100$ give 12/12 successes at $\kappa = 0.3$; $\rho_{\mathrm{cvx}} = 1$ gives 4/12, all with $d = 2$; $\delta = 10^{-1}$ reproduces the baseline iterates exactly (Remark 4); the faster-decaying steps $s = 0.75$ and $s = 1$ give no success at $\kappa = 0.3$, and $s = 1$ also fails in all 12 runs at $\kappa = 1$. The failures at $\kappa = 0.3$ are therefore consistent with a diminishing step that is too small for this constraint scale, rather than with a change in feasibility. These tests vary one parameter at a time on one problem family and do not identify a setting that is best in general. On (47), $\delta = 10^{-1}$ and $\delta = 10^{-3}$ give identical iterates because the pseudoconvex track has a single index, at every start either only F_1 is violated or $v_1 - v_2 \ge 4.79$, and the convex track is satisfied from the second evaluated point on; the δ row of Table 8 is therefore a consistency check, not evidence about the role of δ. The synthetic tuning searches consumed 188 295 optimization evaluations and 10.98 seconds of solver time for SG-dyn, and 147 331 evaluations and 8.27 seconds for SG-const, with the same exclusions as above, in the environment stated below.

Reproducibility. The feature-selection results were saved with Python 3.11.7 and NumPy 1.26.4 on Windows, with one BLAS/OpenMP thread requested. The feature oracles use algebraically compressed quadratic forms; checked differences from the original values and gradients are at most 1.82×10^{-12} and 1.10×10^{-11}, respectively. The solver times reported for the feature instances refer to this compressed implementation. The accompanying package contains only the uncompressed oracles; it reproduces the evaluation counts, while the reported solver times are those of the original compressed implementation. Tables 7 and 8 and the synthetic tuning costs

were produced with Python 3.13.4 and NumPy 2.1.3 on Windows with the default number of BLAS threads; their counts are deterministic, and their solver times are single measurements on that machine. With Python 3.13.4 and NumPy 2.1.3, the package also reproduces exactly every numerical entry of Tables 1–5 (Tables 4 and 5 from unrounded scores) and of Table 6, and the counts quoted in Sections 4.3 and 4.4. The supplementary scripts, raw trajectories, tuning records, witnesses, source hashes and validation instructions accompany the results.

5. Conclusion

DTAP solves normal-box nonlinear split feasibility problems with convex and pseudoconvex components by combining a residual-scaled convex step with a safeguarded pseudoconvex step. Under consistency and continuous differentiability, the whole sequence converges to a feasible point; no regularity modulus is an input, although the analysis uses bounds on a compact set containing the iterates. In the feature-selection application all constraints are convex, and test problems with a nonconvex pseudoconvex constraint exercise the pseudoconvex track. In the experiments, tuned subgradient methods can match or outperform DTAP, whose finite-budget performance depends on the starting point and the step scale. Rates for the full residual $J(x^k)$, beyond the convex-track bound of Remark 5, and guarantees under error-bound conditions remain open.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI use

OpenAI Codex was used during revision for language editing.

Data availability

The datasets are available from the UCI Machine Learning Repository [8]. The accompanying reproducibility package contains the scripts, recorded numerical outputs, and instructions for regenerating the reported tables.

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